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RESEARCH ARTICLE

ECG Measurement System for Vehicle Implementation and Heart Disease Classification Using Machine Learning

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ABSTRACT The 12-lead electrocardiogram (ECG) method can diagnose more cardiovascular disease than the single-lead method, but it is difficult to use in daily life because numerous electrodes must be attached to the body. As an aging society approaches, electrocardiography is expanding its use in daily life, even to the heart disease monitoring system for passengers in vehicles. This study proposes a single-lead ECG measurement system implemented in the steering wheel of the vehicle and a machine learning model to classify the driver's heart health status. For the ECG measurement system, an algorithm to obtain stable ECG signals is proposed along with the measurement hardware. It uses the range and interval of the ECG signal to determine stability under noisy conditions caused by vehicle vibration and the driver movement. To classify four classes of heart diseases (normal, atrial fibrillation, other rhythms, noise), a two-stage machine learning structure is proposed. To train the machine learning models with an optimal feature subset, 188 features were extracted from the single-lead ECG dataset, and a sequential wrapper-type feature selection was conducted. As a result, when the naïve Bayes model using ten features was located in the first step and the support vector machine using 13 features in the second step, the proposed two-stage classification structure returned the best score $F1_{NAO} = 0.7898$ and real-time classification performances (0.86 seconds on average).

INDEX TERMS Atrial fibrillation, ECG, feature selection, electrocardiogram, health care, machine learning, two-stage classification.

I. INTRODUCTION

Arrhythmias, such as atrial fibrillation (AF), is one of the diseases that need more attention in our aging society because the incidence of arrhythmias increases with age, and the death rate increases accordingly [1]. In addition, arrhythmias are also one of the main causes of sudden death among young people [2].

AF causes blood retention in the atrium due to abnormal contractions of the atrium, which causes heart failure and thrombus formation. The generated thrombus can lead to complications such as stroke or thromboembolism, which not only increases the risk of cognitive decline and dementia but also increases patient mortality. In addition, while driving a vehicle, an acute cerebral infarction can lead to unconsciousness, which can threaten the safety of the driver and passengers due to the inability to drive. The subsequent necessity for frequent electrocardiogram (ECG) examinations also increases throughout the rest of the patient's life [3].

ECG is a noninvasive method of reviewing the heart's physiological activity. It can diagnose heart diseases, including arrhythmias, ischemic heart disease, hypertrophy of the left and right atrial ventricles, and electrolyte abnormalities, by graphing the electrical activity of the heart. The ECG

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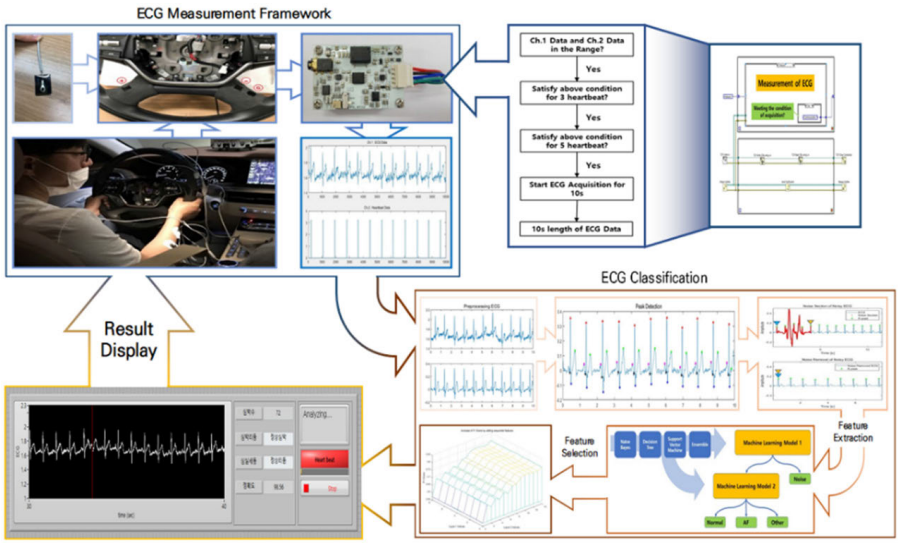


FIGURE 1. Overview of the study.

waveform can be described as a P wave, a QRS wave, and a T wave. The P wave, which comes first, represents the wave of an electrical signal generated in the heart's sinus node. Second, the QRS wave represents an electrical signal that passes through the atrium and spreads to the left and right ventricles until the two ventricles contract. Finally, the T wave is a short rest period of cardiac electrical signals. During the T wave, the heart has a repolarization period as the ventricles are readied for the next heartbeat.

There are several methods of measuring an ECG. The 12-lead ECG measurement method is commonly used in hospitals to obtain an ECG while the patient lies in a stable state. The electrodes are attached not only to the chest, which is close to the patient's heart, but also to the limbs, which are far away from the heart [4]. This enables to obtain accurate results. On the other hand, with the standard limb lead, the method of measuring the ECG using the potential difference between the left and right hands is called a single-lead ECG measurement. It provides convenience and mobility because it requires only two contacts with the body.

If no cardiac abnormalities are detected, or if paroxysmal arrhythmias are suspected during the 12-lead ECG, which takes about 5 to 10 min, an ECG monitoring test may be conducted for an additional 24 hours. During that time, the Holter test [5] is performed, and in some cases, ECG data are acquired using two contact surfaces with the body, such as the patient's hands, mainly using the single-lead ECG measurement method. This allows the patient to perform the Holter test, which is performed intermittently for 24 hours during regular daily activities, regardless of place or time. However, unstable contact with the sensor and excessive movement interferes with obtaining a clean ECG signal. In this circumstance, the 12-lead method can provide more accurate diagnoses than the single-lead method because it is strong to noise and accurately expresses the rise of the ST segment and the T waveform. However, patients cannot easily measure themselves during daily activities because of the inconvenience and mobility issues.

Many state-of-the-art studies have contributed to predicting heart diseases based on single-lead ECGs. For example, a dry electrode was applied to the steering wheel of a vehicle [6]. For this method, a polyester fiber soaked in silver chloride is wrapped around the steering wheel to obtain an ECG through contact between the driver's hands. The proposed dry electrode had excellent signal-to-noise ratio (SNR) and R peak detection rates in various driving conditions. It is also able to detect driver drowsiness by calculating the heart rate using the R peak.

An ECG signal is a type of multivariate time series data, which means it is a series of time-ordered data points associated with multiple time-dependent variables containing local and global patterns [7]. There are also several studies to apply the advanced machine learning technique to time series data [8], [9]. In addition, there are many studies to use artificial intelligence for heart disease inference from ECG data.

In [10], heart disease was predicted using machine learning with an ensemble (ENS) model trained by extracting 74 features from the ECG and selecting 40 optimal features. In addition, the performance of noise classification was improved by using the R peak interval and four features using matrix decomposition.

Another study [11] predicted heart disease using deep learning. Single-lead ECG classification was performed using a restricted Boltzmann machine and deep belief network. Deep learning training uses features extracted from ECG peak points. This study showed excellent accuracy, even with low-resolution data of 114 Hz.

Both deep learning and machine learning were used to predict heart disease [12]. The features extracted from 43 ECGs were used for training the ENS model. The ECGs were

filtered with nine specific frequencies using the Hilbert transform, and then they were used to train the convolutional neural network model. It showed high levels of accuracy when classifying a single-lead ECG using both deep learning and machine learning simultaneously.

Many studies have achieved high accuracy and F1 scores when predicting heart diseases using single-lead ECG data. However, in many cases, only a few datasets were used, or clean ECG data were employed, which do not contain noises. Also, most studies focus on ECG classification using artificial intelligence, and few studies deal with the application or implementation of ECG measurement and classification.

Therefore, this study aims to implement an ECG measurement system in a vehicle to predict the heart diseases of the driver by combining the steering wheel with a single-lead ECG measurement device. This implementation enables ECG measurements in daily life, such as driving situations, and facilitates the prediction of heart diseases, such as AF, using machine learning.

There are some issues to be solved in measuring ECG signals in a vehicle. First, stable and clean ECG data should be obtained during driving. It is not easy to acquire clean ECG data from the driver during driving. Vibration by vehicle movement, hand gripping force to the steering wheel, and even breathing of the driver cause noise to ECG signals. Second, all algorithms, including artificial intelligence, should be computationally low for real-time prediction and vehicle application. A vehicle has a lot of embedded systems, and it is very significant to take low computational resources for cost reduction. The main goals of this study are as follows.

- Development of an algorithm that stably acquires the driver's ECG data by combining a single-lead ECG measurement device with the steering wheel.
- Construction of a two-step machine learning classification structure to improve classification performance for AF and noise.
- Extraction of multiple features from single-lead ECG data with noise and selection of features that are optimized for each step.
- Reduction of computational resources used in performing real-time predictions and improvement in machine learning model classification performance through dimensionality reduction according to feature selection.

An overview of the study is described in Fig. 1. This study mainly involves the acquisition of an ECG from the driver and then the process of classifying the acquired ECG. First, the ECG acquisition chapter presents an algorithm that can acquire a stable ECG signal from the driver by combining a single-lead ECG measurement device with the steering wheel of a vehicle. That is followed by an introduction of the communication between the measurement and analysis tools and the user interface (UI) displayed to the driver.

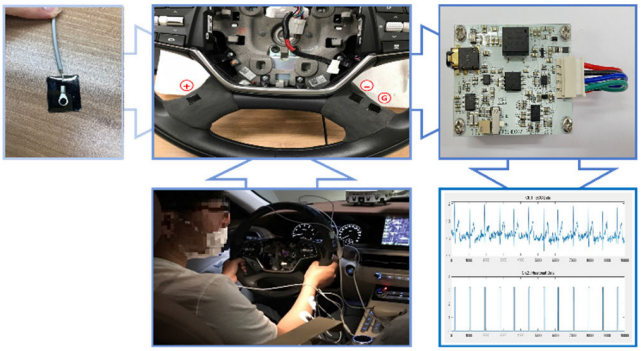


FIGURE 2. ECG acquisition system (a) AgCl metal plate with lead cable (b) Sensors located on the steering wheel (c) ECG signal from channel 1 and heart rate from channel 2.

Second, the ECG classification chapter presents the methods of preprocessing the obtained ECG signal and the removal of the noise. The chapter also introduces the features extracted from single-lead ECG data. Then, a machine learning model with a two-step structure and the performance of a sequential wrapper-type feature selection are introduced.

Finally, the results chapter evaluates the classification and real-time prediction performances of the proposed two-stage structured machine learning model.

II. ECG ACQUISITION

A. SYSTEM INTEGRATION

It is not easy to obtain an ECG signal from a driver in the limited space inside the vehicle. Therefore, by using the single-lead ECG measurement method to acquire signals with only two points of contact with the body, convenience and mobility are maximized for the driver. The sensor was positioned at the points where the driver's hands are in contact with the vehicle's steering wheel during driving.

As a single-lead ECG measuring device, a small two-channel ECG module was used. This module provides a sampling rate of 1,000 Hz and a notch filter of 50/60 Hz to respond to power interference. In addition, a high-pass filter and a low-pass filter with cutoff frequencies of 0.3 Hz and 35 Hz, respectively, were applied to remove noise from the driver breathing in the low-frequency range [13], and the noise generated by muscle movement in the high-frequency range [14].

As shown in Fig. 2(a), Ag/AgCl metal plates connected to each lead cable were utilized as an ECG sensor. These metal sensors were fixed to the steering wheel so that the anode is in contact with the driver's left hand and the cathode and reference are in contact with the right hand, as shown in Fig. 2(b). By placing sensors at the five and seven o'clock positions of the steering wheel, the driver's shoulder muscle tension can be relieved, and noise in the high-frequency region generated from muscle movement can be minimized. Through the ECG module, as in Fig. 2(c), ECG raw signal is acquired along with the heartbeat rate from the signal.

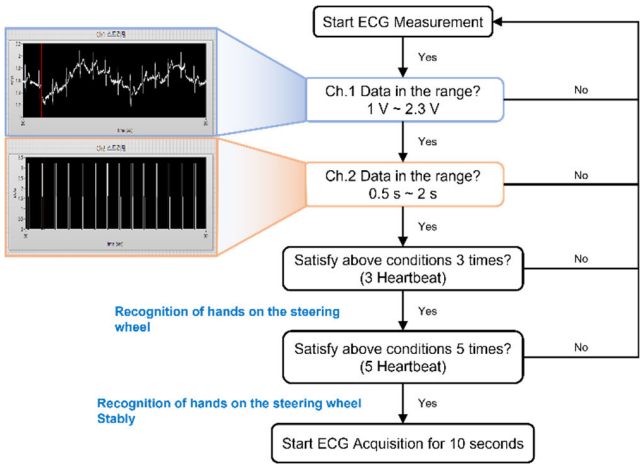
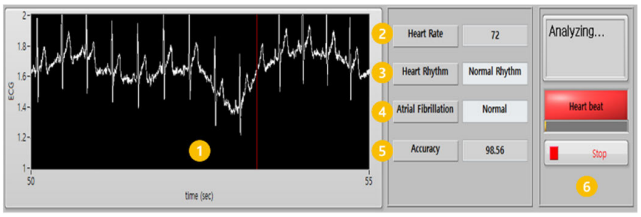


FIGURE 3. ECG acquisition algorithm.



① ECG signal ② Heart rate ③ Heart rate rhythm ④ Predicted class
⑤ Prediction score ⑥ Status window

FIGURE 4. Driver UI panel.

B. ECG MEASUREMENT

For ECG data acquisition from a driver, the driver's hands are gently placed on the steering wheel. The driver's left hand is in contact with the anode, while the right hand is in contact with both cathode and reference. Because only a stable ECG signal should be collected to prevent the wrong prediction of the driver's heart disease, an algorithm to determine whether the ECG signal is stable is necessary.

In this study, LabVIEW was used to collect ECG data. An ECG signal is measured and plotted in the range of 0 V to 3.3 V and a normal heartbeat is measured and plotted between 60 and 100 beats per minute. The algorithm to determine a stable ECG signal in Fig. 3 was developed by fusing these two pieces of information. In this way, noise is excluded so that clean ECG data can be obtained.

With the ECG data acquired through Ch. 1 maintained in the range of 1V–2.3V if the time interval between QRS complex measurements is between 0.5 seconds and 2 seconds (30 bpm to 120 bpm, a more comprehensive range than normal), it acts as a trigger for ECG data accumulation. When the trigger is activated three times, the driver's hands are recognized as being in stable contact with the steering wheel sensor. There is another condition to trigger the ECG data accumulation for a clean ECG signal acquisition. When the first triggering condition is consecutively activated five times, the ECG data through Ch. 1 is accumulated in earnest to constitute the ECG data of 10 seconds in length.

The formed 10-second-long ECG data are transmitted to MATLAB, and then preprocessing is conducted to remove noise. Finally, heart disease prediction is made through feature extraction and machine learning. In addition to the predicted disease, the predicted score for the disease and the heart rate using the R peak of the driver's ECG signal are calculated and transmitted back to LabVIEW for a display to the driver.

C. SYSTEM UI AND COMMUNICATION

The ECG measurement program consists of two while loops. Following the master/slave design pattern using notifier, one master while loop acts as a trigger for ECG measurement accumulation and data transmission/reception through a state machine structure. The slave while loop transmits ECG data through local TCP/IP communication with MATLAB, and receives and displays the results.

As previously mentioned, when the state machine structure is in the measurement state, it follows the flowchart shown in Fig. 3. When the 10-second length of ECG data are accumulated, the state machine structure goes to the data transmission & reception state, converting the stored ECG data into a one-dimension array and sending it to MATLAB. In the measurement state, if the Ch. 1 data representing the ECG is out of the range of 1V–2.3V, or if the Ch. 2 data representing the heart rate interval is out of the range of 0.5–2 seconds, the algorithm starts the ECG data measurement from the beginning again. The ECG data transmitted to MATLAB in a one-dimension form is subjected to two noise reduction methods. The baseline fluctuation noise caused by unstable contact with the sensor and the noise in the high-frequency region generated by muscle movement are attenuated through the filter, and the noise region is deleted using the R peak of the ECG data. Feature extraction is performed using the noise-removed ECG data, and classification into one of four classes is performed through a machine learning model. In addition to the predicted class result, the result array consisting of the driver's heart rate and the speed of the pulse (normal, tachycardia, bradycardia), calculated using the R peak and the predicted score for the class result, is sent back to LabVIEW, and displayed for the driver, as shown in Fig. 4.

III. ECG CLASSIFICATION

A. DATASET

In this study, the 8,528 single-lead ECG data from the PhysioNet Computing in Cardiology Challenge 2017 were used to train and validate machine learning classification models [15]. The ECG data were measured with an AliveCor device sampling at 300 Hz and consist of time series data with a minimum length of 9 seconds and a maximum length of 61 seconds. Each ECG was placed in one of four classes: normal ECGs (N; 5,076), AF ECGs (A; 758), other diseases ECGs (O; 2,415), and noisy ECGs (∼, 279), as shown in Fig. 5. Table 1 summarizes the dataset.

TABLE 1. Physionet 2017 challenge dataset.

Type (Label)	Number	Time Length(s)				
		Mean	SD	Max	Median	Min
Normal (N)	5,076	32.11	9.97	60.95	30	9.05
AF (A)	758	32.34	12.32	60.21	30	9.99
Other rhythm (O)	2,415	34.30	11.76	60.86	30	9.13
Noisy (~)	279	24.39	10.43	60	30	9.36
Total	8,528	32.50	10.89	60.95	30	9.05

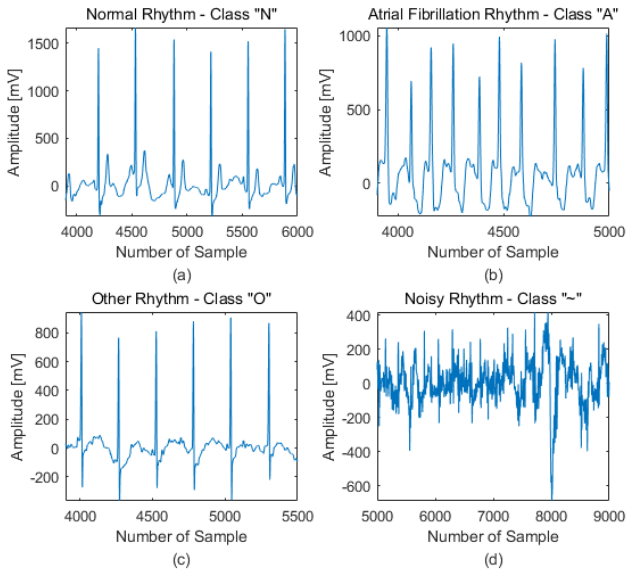


FIGURE 5. ECG data with four classes (a) Normal (b) AF (c) Other (d) Noisy.

The ECGs with other diseases include the following [16]:

- Tachycardia (mean heart rate over 100 bpm);
- Bradycardia (mean heart rate under 50 bpm);
- Wide QRS complex (longer than 110 milliseconds);
- Presence of ventricular or fusion beats;
- Presence of at least one extrasystole;
- Long PR interval (longer than 210 milliseconds);
- Ventricular tachycardia;
- Atrial flutter.

Additionally, there are ECG data included, which had been turned upside down due to the sensors placed in the wrong positions. Because the R peak values for these data have negative values, these data can be detected and were inverted using the average and median values of the R peak amplitude for the entire length of the ECG data.

There are two differences between the training data and the ECG data measured by the ECG sensor module on the steering wheel. The first difference is that the training data had a sampling rate of 300 Hz, whereas the ECGs acquired from the steering wheel had a sampling rate of 1,000 Hz. The higher the sampling rate, the higher the possibility of accurate feature extraction from the ECG data. By using a low-pass filter [17], 300 Hz training data were up-sampled to 1,000 Hz by correcting the delay caused by the filter.

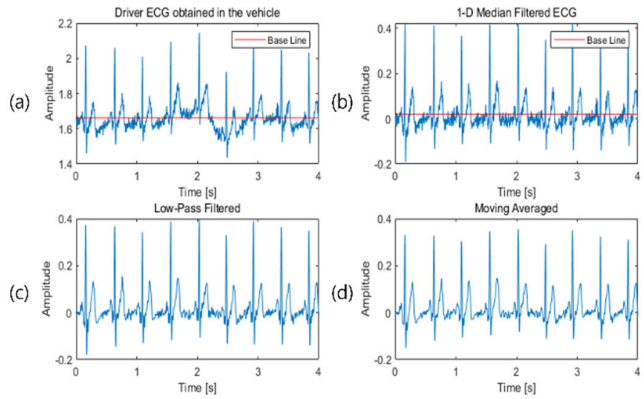


FIGURE 6. Baseline fluctuation removal and high-frequency region noise cancellation (a) Original ECG signal from the driver (b) One-dimension median filtered ECG (c) One-dimension median filtered and low-pass filtered ECG (d) One-dimension median filtered, low-pass filtered, and moving averaged ECG.

B. PREPROCESSING ECG

Acquiring the driver's ECG data using the single-lead ECG measurement is not free from various noise sources. During driving, vibration from vehicle movement, breathing of the driver [13], electromyography (EMG) signal, and surface resistance change when gripping the steering wheel sensor generates noise in the high-frequency range and baseline fluctuation noise [14].

Therefore, the driver's ECG data need to be refined using a digital filter. In this study, baseline fluctuation noise removal was performed with a median filter [18], and then noise removal from the EMG signal in the high-frequency range was performed using a low-pass filter and a moving average. For the median filter, third order of the one-dimensional median filter was employed. For the low pass filter, the pass band frequency is 0.045π radians/sample and the stop band attenuation is 120 dB, and a sliding window of 20 ms was employed for the moving average. Fig. 6 shows the preprocessing of ECG.

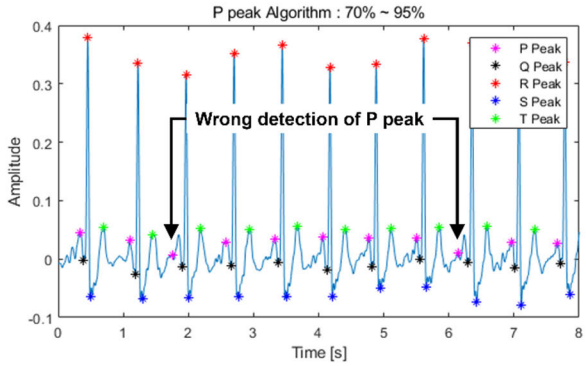


FIGURE 7. Wrong detection of P peaks by algorithm.

C. PEAK DETECTION

Most of the features used to train the machine learning classification models are extracted based on the PQRST waves. The P wave represents the depolarization of the right atrium on the ECG, the QRS complex represents the depolarization state of the ventricle, and the T wave represents the normal repolarization of the ventricle. It is important to locate the correct locations of the P, Q, R, S, and T peak points in the ECG data. In particular, the R peak readily stands out because it has a larger amplitude than other peaks, and it is relatively easy to detect it even with disturbances caused by various noises. Therefore, after extracting the R peak first, other peaks can be derived based on the R peak locations. To do this, the R peak was detected first using an open source Windows peak detector [19] based on the Pan–Tompkins [20] algorithm, and then the other peak points were derived using the R peak as a reference. The difference of the method introduced in this study from the existing Pan–Tompkins algorithm is that the band-pass filter is replaced with a newly designed Mexican hat filter.

Based on the derived R peak, the remaining P, Q, S, and T peak points are detected. In [21], an algorithm was constructed for the detection of the P, Q, S, and T peaks. The algorithm uses the distances from each peak point in the R peak interval after a cardiologist directly marks the P, Q, S, and T peak points. The criteria are as follows:

- For each R peak, the maximum value that exists at 70% to 95% of the distance from the previous R peak is detected as the P peak.
- For each R peak, the minimum value located at 80% to 100% of the distance from the previous R peak is detected as the Q peak.
- For each R peak, the minimum value is detected as the S peak, and the maximum value is detected as the T peak from the extreme values at positions 0% to 60% of the distance to the subsequent R peak.

When peak detection was performed on the ECG data up-sampled from 300 Hz to 1,000 Hz using the presented algorithm, the R peaks were still displayed at accurate positions, whereas the P peak detection was somewhat inaccurate, as shown in Fig. 7.

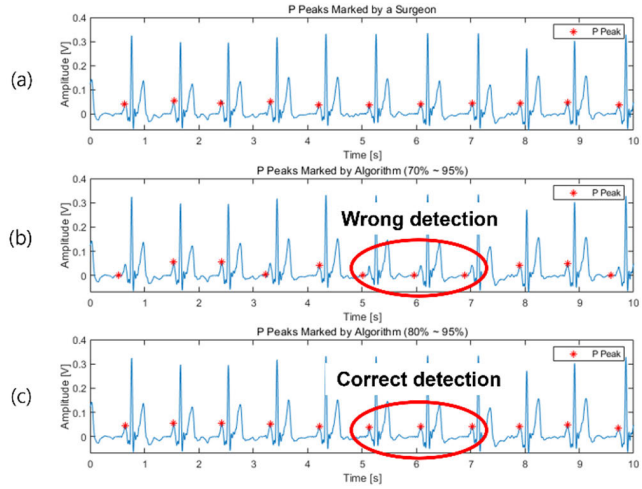


FIGURE 8. Comparison of P peak detection (a) P peak detection by a surgeon (b) P peak detection by algorithm (70%–95%) (c) P peak detection by algorithm (80%–95%).

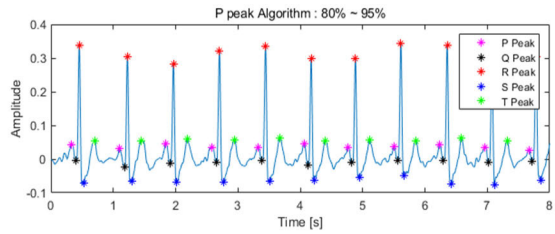


FIGURE 9. Improvement of peak detection.

In order to improve the P peak detection accuracy, 20 ECGs were randomly selected for each of the classes except the noise class. For the selected ECG data, with the help of a surgeon, the exact P peak locations were marked as shown in Fig. 8(a).

Based on the exact P peak locations, the existing P peak detection algorithm was modified. The existing algorithm that had detected the P peak within 70% to 95% of the distance from the previous R peak was modified to detect the P peak in the range of 80% to 95% of the distance, as shown in Fig. 8(b) and (c). When the P peak value errors detected using the algorithm before and after correction were expressed as the root mean square error, the error rate decreased from 60.2 to 31.4 for the AF class, 57.05 to 19.95 for the normal class, and from 84.26 to 38.15 for the other diseases class. In addition, the error rate was greatly reduced when the difference between the P peak detected by the algorithm before and after correction and the P peak manually marked by the surgeon was expressed in time, as shown in Fig. 8. Peak detection after modifying the algorithm is shown in Fig. 9.

D. NOISE REMOVAL USING R PEAKS

The driver's ECG data measured by the sensor on the vehicle's steering wheel are acquired through the single-lead measurement method, which is seriously disturbed by the driver movement and breathing. This can result in completely meaningless signals being included in driver ECG input, and

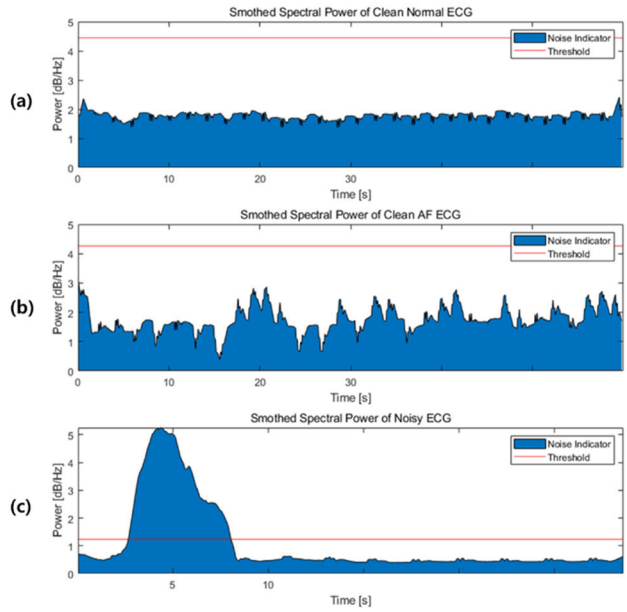


FIGURE 10. Spectrogram and calculated threshold for each ECG class to determine the noise area (a) Smoothed spectrogram of a normal ECG (b) Smoothed spectrogram of an AF ECG (c) Smoothed spectrogram of a noisy ECG.

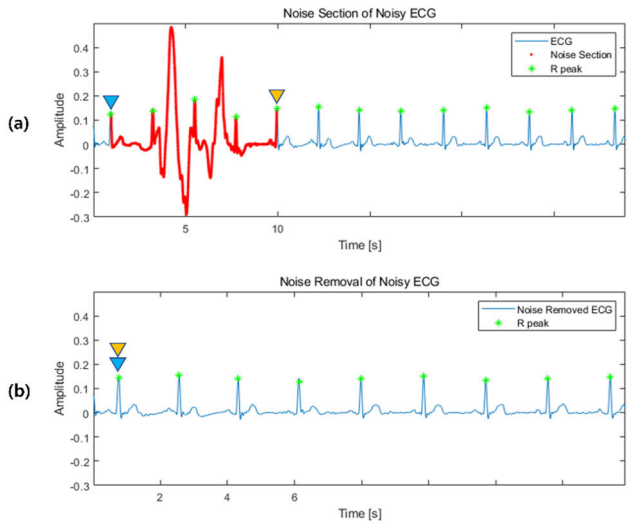


FIGURE 11. Removal of noise region (a) Noise region (red mark) (b) ECG signal after noise removal.

this may result in the degradation of the machine learning classifier's performance due to poor feature extraction and prediction.

To solve this, noise reduction was performed with a spectrogram [21]. First, short-time Fourier-transform (STFT) was performed on the complete ECG data from 0 Hz to 500 Hz, using a 100-ms-long window with an overlap length of 90 ms. After STFT, a moving average was performed with a window of 0.1 ms in the high-frequency region, that is, the spectrum between 100 Hz and 500 Hz. The normal ECG had an almost constant shape, as shown in Fig. 10(a), but the AF ECG showed some sharpness due to the irregular heartbeat, as shown in Fig. 10(b). Also, for the ECG containing noise,

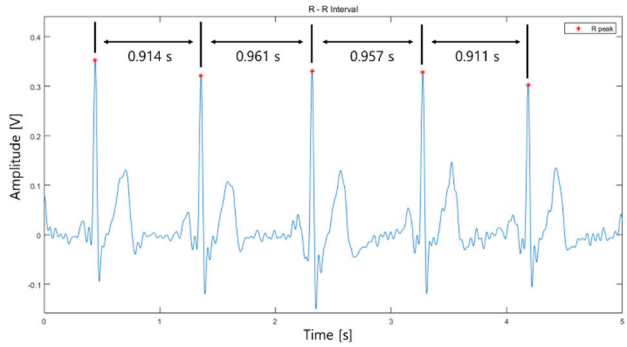


FIGURE 12. R peak interval of AF ECG.

a prominently protruding shape can be found, as shown in Fig. 10(c). Because there may be one or more noise regions for each ECG or within any one ECG, a spectral threshold that is capable of discriminating noise was empirically calculated and presented [21]. A value 2.5 times the median of the moving average performed for the spectrum in the high-frequency region was set as the threshold, and the region exceeding this value was recognized as noise. Then, the R peak interval that included noise was removed by using the R peak point just before the noise started and the R peak point after the noise ended, as shown in Fig. 11(a) and (b).

E. FEATURE EXTRACTION

The normal ECG can be expressed as a P wave indicating the contraction of the atrium, a QRS complex indicating the contraction of the ventricle, and a T wave indicating the relaxation of the ventricle. The main characteristics of arrhythmias such as AF are an irregular heartbeat and the absence of P waves. In particular, through the R peak of the ECG, features such as changes in heart rate can be easily found, as shown in Fig. 12.

Because the R peak has the sharpest and largest amplitude compared to other peak points, it is more resistant to noise generated during the driver's ECG acquisition using the single-lead ECG method. In this regard, various features using R peaks, such as the number of R peaks and the range and changes of R peak intervals, are extracted. In addition, morphological and statistical features using the remaining peak points are extracted. Features such as the ratio of the number of R peaks to the number of P peaks, the range of the interval between R and P peaks, and the slope of the QR segment and RS segment are shown in Table 2.

The QT interval is used as an indicator for arrhythmia detection, but the QT interval also changes with the R peak interval because the heart rate changes. That is, as the heart rate increases, the QT interval decreases. In order to correct the QT interval (which varies according to heart rate) to a value that does not depend on heart rate, (1), (2), and (3) were also derived as features [23], [24], [25].

$$Bazett's\ fomula = \frac{QT}{\sqrt{\frac{RR}{1s}}} \tag{1}$$

TABLE 2. Extracted Morphological and Statistical Features.

Morphological and Statistical Features
Coefficient of variation of RR and dRR [22]
Mean, median, min, max, skewness, kurtosis, range, variance of RR intervals
Coefficient of variation, range, median of S peak over R peak
Median, coefficient of variation, number of negative slopes of ST segment
Number of S peak amplitude less than corresponding Q peak amplitude
Median distance of ST segment higher than corresponding QT peak segment
Median value of T peak amplitude over R peak amplitude
P peak count over R peak count
Median, coefficient of variation of QR segment and QS segment
Median, coefficient of variation, range of Q peak amplitude over R peak amplitude
Median, coefficient of variation of S peak depth over R peak height using Q peak as a reference
Median, coefficient of variation of QR segment and RS segment slope
Median, coefficient of variation of slope between S peak and 50 ms after that S peak
Standard deviation of R peak interval, and P peak interval and ratio of count of P peak interval and R peak interval
Standard deviation of P peak and R peak intervals except for outliers greater than 120 ms
Median of R peak interval subtracted from median of P peak interval
Median, range, number of R peak interval and P peak interval
Ratio of the front median to the back median as the midpoint of the R peak interval

$$Fridericia's\ fomula = \frac{QT}{\sqrt[3]{\frac{RR}{1s}}} \tag{2}$$

$$Sagie's\ fomula = 1000\left(\frac{QT}{1000} + 0.154\,(1 - RR)\right) \tag{3}$$

R peak interval and pictorial features were also extracted. In [26], evidence was presented using the change in R peak interval and Lorenz plot [27]. The direction of the change in the R peak interval was drawn by expressing the change in the R peak interval at a previous point with respect to the change in the R peak interval at a specific point. Lorenz plot was expressed numerically through a two-dimension histogram, and features were extracted using points drawn on a plane divided into 12 regions, as shown in Fig. 13(a).

BinCount indicates the number of bins constituting the histogram, and PointCount indicates the number of points belonging to the bin. The following features are derived with the BinCount and PointCount.

$$IrregularityEvidence = \sum_{n=1}^{12} BiNCount_n \tag{4}$$

$$DensityEvidence = \sum_{n=5}^{12} (PointCount_n - BinCount_n) \tag{5}$$

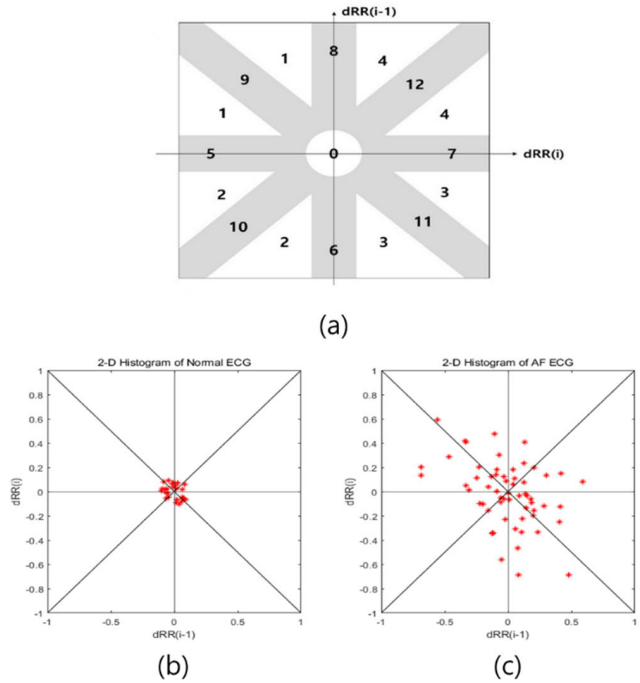


FIGURE 13. Characteristics of normal and AF ECGs in Lorenz plot (a) Twelve areas of the Lorenz plot (b) Two-dimension histogram of normal ECG (c) Two-dimension histogram of AF ECG.

$$AnisotropyEvidence = \left|\sum_{n=9,11} PointCount_n - \sum_{n=10,12} PointCount_n\right| + \left|\sum_{n=6,7} PointCount_n - \sum_{n=5,8} PointCount_n\right| \tag{6}$$

$$PACEvidence = \sum_{n=1}^{4}(PointCount_n - BinCount_n) + \sum_{n=5,6,10}(PointCount_n - BinCount_n) \tag{7}$$

$$AFEvidence = IrregularityEvidence - OriginCount - 2*PACEvidence \tag{8}$$

$$ATEvidence = IrregularityEvidence + AnisotropyEvidence + DensityEvidence + RegularityEvidence - 4 * PACEvidence \tag{9}$$

$$OrgIndex = OriginCount + AnisotropyEvidence$$

$$+ DensityEvidence + RegularityEvidence - 2 * IrrgularityEvidence \quad (10)$$

Irregularity indicates the sparseness of the distribution on the two-dimension histogram. A lower value indicates normal sinus rhythm, as shown in Fig. 13(b), and a higher value indicates AF, as shown in Fig. 13(c). DensityEvidence is clustering density, AnisotropyEvidence is the distribution direction, and PACEvidence is the compensatory pauses in the two-dimension histogram. AFEvidence and ATEvidence are calculated from the combination of these variables and are used to discriminate between AF and atrial tachycardia.

Next, [28] also extracted features from the visual aspect using the Poincaré plot. After applying a discrete wavelet transform to ECG data to remove baseline wandering and detect the QRS complex, feature extraction using inter-beat intervals was performed. Using the interval between consecutive QRS complexes obtained from one given ECG, the current interval for the previous interval was expressed in a two-dimension Poincaré plot, and three features were extracted. First was a feature using the distance between points on the Poincaré plot, which means the rate of change of inter-beat intervals and it is calculated with the following equation.

$$\textit{Mean Stepping increment of inter beat intervals} = \frac{\frac{1}{n-2}\sum_{j=1}^{n-2}\sqrt{\left(I_j - I_{j+1}\right)^2 + \left(I_{j+1} - I_{j+2}\right)^2}}{\frac{1}{n}\sum_{j=1}^{n} I_j} \quad (11)$$

Here, I_j means the jth interval among all n_{th} R peak intervals and the unit is ms. Next, a point (a, a) that minimizes the sum of the distances between points on the Poincaré plot and a point on $y = x$ were derived to indicate how spread out around the diagonal line the points on the Poincaré plot are using the formula

$$(a, a) = \frac{1}{2(n-1)}\left(-I_1 - I_n + 2\sum_{j=1}^{n-1} I_j\right) \quad (12)$$

The third is a feature using the number of clusters formed by clustering points on the Poincaré plot based on spectral graph theory [29]. In the case of a normal ECG, a small number of clusters are formed as shown in Fig. 14(a), whereas in the case of AF, one or innumerable clusters are formed as shown in Fig. 14(b).

In addition, features in terms of heart rate variability (HRV) and frequency were also extracted, forming a total of about 180 feature groups as shown in Table 3.

F. COMPOSITION OF TWO-STAGE CLASSIFICATION STRUCTURE AND FEATURE SELECTION

The single-lead ECG method enables ECG testing using only two contact points with the body, but the movement of the vehicle and driver or an unstable contact with the sensor inevitably generates noise in the ECG signal. Therefore, the preprocessing procedure introduced in Chapter III-B and noise region removal using the R peak introduced in Chapter III-D are used to alleviate the noise effect on the collected ECG signal.

In terms of the ECG signal characteristics, as described in Chapter III-E, most features are related to the ECG peak points. In addition, the peak points of the noisy ECG data show irregular and inconsistent patterns, unlike the peak points of the clean ECG data, as shown in Fig. 15.

Paying attention to these obvious differences, this study introduces a two-stage classification structure, as shown in Fig. 16. In the first step, classification is carried out on the noise class. In the second step, classification is performed on the remaining three classes (normal, AF, other).

TABLE 3. Extracted features of frequency and HRV domain.

Frequency and HRV Domain Features
HRV features (pNN20, pNN50, standard deviation of successive RR intervals) [30, 31]
Shannon, Tsallis, Renyi entropies, and tenth order linear prediction filter coefficients and wavelet transform based features [32]
Spectral centroid, 85% roll-off, flux, kurtosis of ECG frequency truncated to 2-second window
Welch's power spectral density estimates of R peak interval
10% Trimmed mean; skewness, 80% domain; kurtosis between 0.5 and 5 Hz of ECG frequency domain; and kurtosis of ECG time domain
Signal-to-noise ratio, total harmonic distortion, zero-crossing rate of ECG time series
Width of a particular frequency domain relative to the overall frequency domain using Welch power spectrum [33]
Features using sample entropy, approximate entropy, detrended fluctuation analysis, Poincaré plot geometry [34-36]
Features for extreme bradycardia, extreme tachycardia, asystole, ventricular tachycardia, and ventricular fibrillation classification [37]
Signal quality index related features [38]
Spectral purity index [39-41]
Adaptive mathematical morphology for QRS fiducial points detection [42]

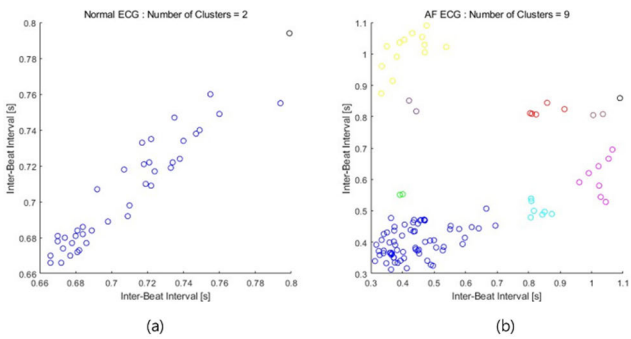


FIGURE 14. Clustering using R peak shifts in normal and AF ECGs (a) Clustering of inter-beat intervals of normal ECG (b) Clustering of inter-beat intervals of AF ECG.

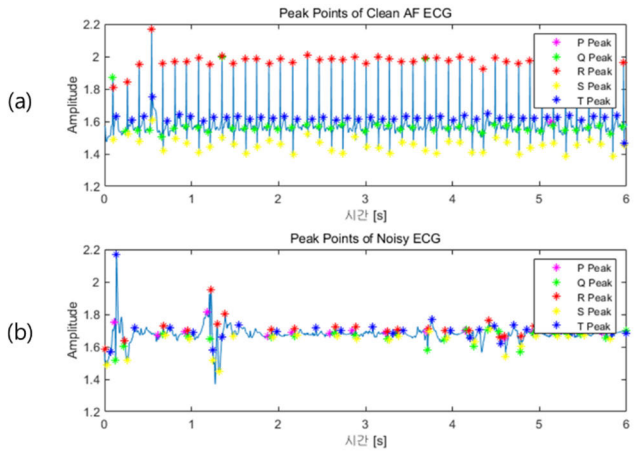


FIGURE 15. Characteristics in ECG peak points (a) Clean ECG (b) Noisy ECG.

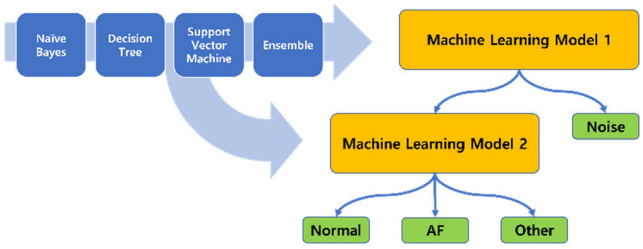


FIGURE 16. Two-stage classification structure using four machine learning models.

Four types of machine learning classification models—Naïve Bayes (NB), Decision Tree (DT), Support Vector Machine (SVM), and Ensemble (ENS)—were considered for the two-level structure. In order to find the feature subset in which the four classification models could perform optimally at each stage, 80% of training dataset of 8,528 items and forward sequential wrapper-type feature selection [43] through fivefold cross-validation were performed.

For example, considering the NB model in the first stage of performing classification on the noise class and the other classes, by sequentially adding about 180 extracted features one by one, the NB model was trained and evaluated. In addition, the feature subset that can classify the noise class and the other classes with the best performance was investigated. In this way, feature selection was performed for eight cases, where four machine learning models were considered in two steps. After that, training through the selected feature subset and hyperparameter optimization for each model was carried out to evaluate the remaining 20% of the test data for a total of 16 cases that can be composed of a two-step structure and four classification models. Then, a classification model was selected for each node.

IV. RESULTS

A. EVALUATION METHOD

Accuracy and F1 score are employed for performance metrics. Accuracy is a performance metric of the prediction accuracy of a machine learning model. It simply measures the percentage of correct predictions with the following equation.

TABLE 4. Counting rules for the numbers of variables.

		Predicted Classification				
		Normal	AF	Other	Noisy	Total
Reference Classification	Normal	N_n	N_a	N_o	N_p	$\sum N$
	AF	A_n	A_a	A_o	A_p	$\sum A$
	Other	O_n	O_a	O_o	O_p	$\sum O$
	Noisy	P_n	P_a	P_o	P_p	$\sum P$
	Total	$\sum n$	$\sum a$	$\sum o$	$\sum p$	

$$\begin{aligned} TP &: True\ Positive \\ TN &: True\ Negative \\ FP &: False\ Positive \\ FN &: False\ Negative \\ Accuracy &= \frac{TP+TN}{TP+TN+FP+FN} \end{aligned} \tag{13}$$

However, it is difficult to use as a performance comparison index when there is an unbalanced amount of data between classes, as with the PhysioNet 2017 Challenge dataset. Therefore, the F1 score is also employed for performance metrics as in Table 4 and the following equations.

$$F1\ Score\ of\ Normal\ Rhythm : F1_N = \frac{2 * N_n}{\sum N + \sum n} \tag{14}$$

$$F1\ Score\ of\ AF\ Rhythm : F1_A = \frac{2 * A_a}{\sum A + \sum a} \tag{15}$$

$$F1\ Score\ of\ Other\ Rhythm : F1_O = \frac{2 * O_o}{\sum O + \sum o} \tag{16}$$

$$F1\ Score\ of\ Noisy : F1_P = \frac{2 * P_p}{\sum P + \sum p} \tag{17}$$

$$Total\ F1\ Score\ :\ F1_{NAOP} = \frac{F1_N + F1_A + F1_O + F1_P}{4} \tag{18}$$

In addition, because there was minimal data in the noise class, the F1 score ($F1_P$) for the noise class was excluded from the total F1 score ($F1_{NAOP}$), and performance was evaluated with the following formula.

$$F1_{NAO} = \frac{F1_N + F1_A + F1_O}{3} \tag{19}$$

B. EVALUATION OF MACHINE LEARNING MODEL AND TWO-STAGE CLASSIFICATION STRUCTURE

In the first step of discriminating noise class from non-noise classes (normal, AF, other), ten optimal features were selected for NB, one for DT, one for SVM, and one for

ENS. With the selected optimal features, each model was trained through fivefold cross-validation and hyperparameter optimization to minimize cross-validation loss. As a result of the first step, the Epanechnikov smoothing method with a window width of 0.12 was adopted for the NB model. For the DT model, MinLeafSize = 1, MaxNumSplits = 1, and SplitCriterion = deviance were adopted. For the SVM model, BoxConstraint = 2.559 and KernelFunction = linear were adopted. The ENS model adopted GentleBoost, MinLeafSize = 40, and MaxNumSplits = 5640. The trained models showed 97.18%, 96.66%, 96.66%, and 96.66% accuracy, respectively, for 20% of the test data in the first step.

In the second step, classifying normal, AF, and other, 12 optimal features were selected for NB, 5 for DT, 13 for SVM, and 13 for ENS. As in the first step, each model was trained through fivefold cross-validation and hyperparameter optimization using the selected optimal features. As a result, the NB model adopted a normal Gaussian distribution for data modeling. For the DT model, MinLeafSize = 11, MaxNumSplits = 3,489, and SplitCriterion = Gini-Simpson diversity index were adopted. The SVM model is a multi-class classifier and has three binary classifiers following the one-versus-all design. It also has a quadratic polynomial as KernelFunction. In the case of the ENS model, the bag method was followed, and twoing was adopted as SplitCriterion, MinaLeafSize = 5, and MaxNumSplits = 5,115. For the second step, the accuracies with the 20% test data by the trained models were 77.45%, 77.09%, 83.15%, and 83.52%, respectively. Table 5 presents the summary.

C. SELECTED FEATURES

NB, DT, SVM, and ENS are considered machine learning classification models of the two-stage structure, and there are a total of 16 two-stage classification structures that can be composed of these four types of models. At each stage, after feature selection, learning is performed using hyperparameter optimization based on the selected features so that each machine learning classification model can provide optimal performance. At each step, feature selection is carried out by training and evaluating sequentially increasing subsets of candidate features that allow each machine learning classification model to achieve the most optimal performance. This process maintains the ratio for each class and uses 80% of the 8,500 data entries as training data, with the remaining 20% as test data, and both are completed through fivefold cross-validation. As a result, the highest F1 score (0.7898) was achieved when the NB binary classifier was located in the first stage, and the SVM multi-class classifier was located in the second stage. In the first binary NB classifier, the following ten features were selected:

- Anisotropy evidence [26];
- Number of clusters in Poincaré plot using R peaks [28];
- Minimum interval of R peaks;
- Median value of S peak to R peak ratio;
- Number of S peaks under Q peaks;

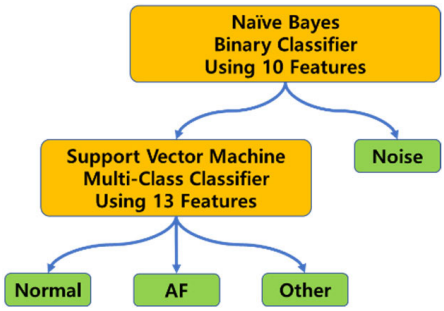


FIGURE 17. Machine learning models with features.

- Median value of S peak depth over R peak height using the Q peak as a reference;
- Coefficient variance of QR segment slope;
- Number of P peak interval to number of R peak interval ratio;
- Number of R peak intervals;
- Maximum location of sample entropy using R peak intervals [36].

Next, in the multi-class SVM classifier, the following 13 features were selected:

- AF evidence [26];
- Irregularity evidence [26];
- Skewness of RR intervals;
- Shannon entropy of RR intervals [32];
- Complexity;
- Coefficient variance of R peak amplitude;
- Number of P peak intervals;
- Detrended fluctuation analysis of sample entropy using R peak intervals [36];
- Asystole evidence [37];
- Ventricular tachycardia, ventricular fibrillation evidence [39];
- Relative power in the baseline of the signal quality index [38];
- Minimum heart rate of five consecutive beats.

Fig. 17 shows the adopted machine learning model and the number of selected features at each stage. Fig. 18 shows the rising F1 score when the adopted machine learning model with the selected additional features is trained. Table 6 provides the summary.

D. REAL-TIME ADAPTABILITY

A 10-second-long ECG is obtained from the driver through a single-lead ECG device, and for vehicle implementation of the ECG measurement system, it must take a short time until the system displays a result. For prediction accuracy, the feature subset was investigated to derive the optimal performance for the four types of machine learning classification models (NB, DT, SVM, ENS) and the NB binary classifier and ten feature subsets were adopted in the first step to distinguish noise from normal, AF, and other, and the SVM multi-class classifier and 13 features were used in the

TABLE 5. Performance of machine learning model at each stage.

First Stage: Noise vs. Normal, AF, Other				Second Stage: Normal vs. AF vs. Other			
Model	Number of selected features	Parameters	Acc (%)	Model	Number of selected features	Parameters	Acc (%)
NB	10	Data modeling: Kernel density estimation (Epanechnikov) Window width = 0.12	97.18	NB	12	Data modeling: Kernel density estimation (Gaussian)	76.00
DT	1	MinLeafSize = 1, MaxNumSplits = 1, SplitCriterion = deviance	96.66	DT	5	MinLeafSize = 11 MaxNumSplits = 3,489 SplitCriterion = Gini-Simpson Diversity Index	79.27
SVM	1	BoxConstraint = 2.559, KernelFunction = linear	96.66	SVM	13	BoxConstraint = 0.026 KernelFunction = polynomial (2nd)	84.24
ENS	1	Method = GentleBoost, LearnRate = 0.348, MinLeafSize = 40, MaxNumSplits = 5,640	96.66	ENS	13	Method = bag MinLeafSize = 5 MaxNumSplits = 5,115 SplitCriterion = twoing	82.48

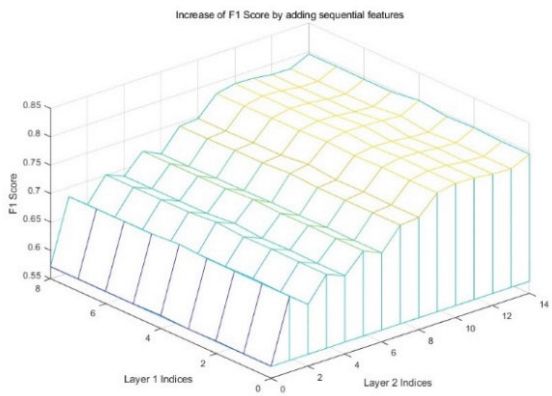


FIGURE 18. Improvement of F1 score with selected features.

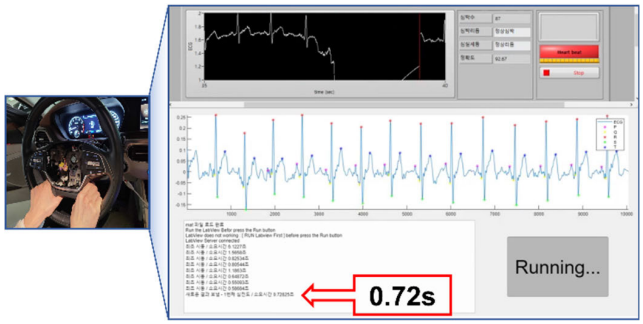


FIGURE 19. ECG acquisition and classification in the vehicle.

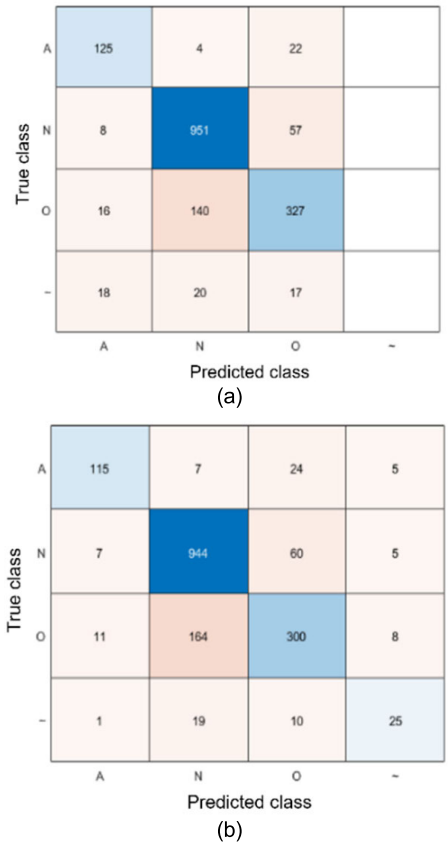


FIGURE 20. Comparison of classification performance using all features and optimal feature subset (a) Confusion matrix with all features (b) Confusion matrix with the feature subset.

second step. The adoption of a subset is intended to reduce the amount of computation resources due to dimensionality reduction and to improve classification accuracy as well as real-time prediction performance.

First, the 10-second ECG obtained from the driver was preprocessed, then the time until noise region removal, feature extraction, and classification into one class was compared. LabVIEW and MATLAB were used to acquire an ECG from the driver and analyze the acquired ECG. These two programs run on a single laptop computer and exchange ECG data and analysis results through local TCP/IP communication. The program operating environment is Intel(R) Core(TM) i7-8700 CPU @ 3.2 GHz processor and 16.0 GB RAM.

Regarding the prediction time, there is a significant time difference between all features and subset features. It takes an average of 2.6 seconds to predict when a total of 180 features are used, whereas it takes an average of 0.7 seconds to predict

TABLE 6. Performances of combinations of machine learning models.

	#1 NB→NB	#2 NB→DT	#3 NB→SVM	#4 NB→ENS	#5 DT→NB	#6 DT→DT	#7 DT→SVM	#8 DT→ENS
Accuracy	0.7578	0.7531	0.8117	0.8158	0.7496	0.7460	0.8047	0.8088
$F1_N$	0.8471	0.8407	0.8781	0.8859	0.8454	0.8393	0.8756	0.8858
$F1_A$	0.6923	0.7088	0.8070	0.7603	0.6711	0.7003	0.7959	0.7500
$F1_O$	0.6019	0.5845	0.6842	0.7034	0.6043	0.5877	0.6861	0.7005
$F1_P$	0.5102	0.5102	0.5102	0.5102	0.0656	0.0656	0.0656	0.0656
$F1_{Total}$	0.6629	0.6610	0.7199	0.7150	0.5466	0.5482	0.6058	0.6005
$F1_{NAO}$	0.7138	0.7113	0.7898	0.7832	0.7069	0.7091	0.7859	0.7788
Normal Accuracy	0.8750	0.8829	0.9291	0.9213	0.8770	0.8839	0.9321	0.9242
AF Accuracy	0.6556	0.6689	0.7616	0.7351	0.6689	0.6887	0.7748	0.7550
Other Accuracy	0.5776	0.5404	0.6211	0.6605	0.5880	0.5549	0.6335	0.6708
Noise Accuracy	0.4545	0.4545	0.4545	0.4545	0.0364	0.0364	0.0364	0.0364
	#9 SVM→NB	#10 SVM→DT	#11 SVM →SVM	#12 SVM →ENS	#13 ENS →NB	#14 ENS →DT	#15 ENS →SVM	#16 ENS →ENS
Accuracy	0.7367	0.7683	0.8152	0.7994	0.7367	0.7683	0.8152	0.7994
$F1_N$	0.8385	0.8534	0.8893	0.8810	0.8385	0.8534	0.8893	0.8810
$F1_A$	0.6867	0.7148	0.7723	0.7362	0.6867	0.7148	0.7723	0.7362
$F1_O$	0.5027	0.6398	0.7050	0.6784	0.5027	0.6398	0.7050	0.6784
$F1_P$	0.0656	0.0656	0.0656	0.0656	0.0656	0.0656	0.0656	0.0656
$F1_{Total}$	0.5234	0.5684	0.6080	0.5903	0.5234	0.5684	0.6080	0.5903
$F1_{NAO}$	0.6760	0.7360	0.7889	0.7652	0.6760	0.7360	0.7889	0.7652
Normal Accuracy	0.9380	0.8878	0.9370	0.9252	0.9380	0.8878	0.9370	0.9252
AF Accuracy	0.7550	0.6887	0.7748	0.7483	0.7550	0.6887	0.7748	0.7483
Other Accuracy	0.3872	0.6253	0.6605	0.6377	0.3872	0.6253	0.6605	0.6377
Noise Accuracy	0.0364	0.0364	0.0364	0.0364	0.0364	0.0364	0.0364	0.0364

when the reduced feature subset is used, as shown in Fig. 20. In addition, the classification accuracy between all features and subset features was also compared using the test data from PhysioNet 2017 dataset. As shown in Fig. 21, they show similar accuracy in predicting the heart diseases of the driver. Considering the prediction time difference, the model with the feature subset can be considered because it is computationally inexpensive.

V. CONCLUSION

This study proposes a single-lead ECG measurement system on the steering wheel that can obtain ECG data from the driver in noisy environment. In order to overcome this noisy environment, an algorithm was also proposed, which determines whether the driver had a stable grip on the steering wheel, with only the ECG and heart rate signal.

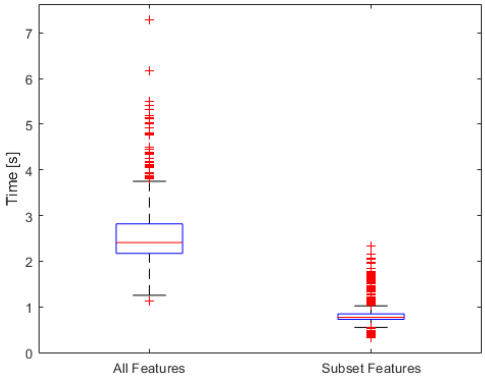


FIGURE 21. Prediction time comparison between all features and subset features.

In addition, by constructing a two-step machine learning classification structure, accurate and real-time heart disease classification was performed with a relatively short 10-second ECG obtained from the driver. Sequential wrapper-type feature selection was performed on the machine learning models (NB, DT, SVM, ENS) to find an optimal feature subset at each stage that can elicit the best performance among a total of 180 features. This resulted in a reduction of unnecessary computing resources and real-time prediction. As a result, a two-stage classification structure was constructed with the NB model using ten features and the SVM using 13 features, which showed excellent performance with F1_NAO=0.7898and F1_A=0.8070. In addition, it took an average of 0.48 seconds to derive the noise class through the two-step classification structure, and an average of 0.86 seconds to derive the normal, AF, and other disease classes, demonstrating real-time prediction performance.

In actual driving, it is very difficult to collect a clean 10-second-long ECG signal from the driver because the signal is always exposed to the artifacts caused by vehicle vibration, bumpy roads, and the steering gripping force of the driver. Therefore, a further study will aim to develop an algorithm that stitches the short length ECG data obtained intermittently whenever the driver's hands are in contact with the sensor on the steering wheel and combines them into one complete 10-second-long ECG signal.

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